Department of Computer Science and Engineering(CSE)

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

Course code:CSE-3524
Course Title:Microprocessor & Microcontroller Sessional

EXPERIMENT 5

Experiment Name: Realization of jump and loop instructions.

OBJECTIVES:

1. To realize the operation of a series using jump instruction.
2. To realize the operation of a series using loop instruction.
3. To check the register values in kit box.
4. To analyze the result.

NECESSARY EQUIPMENT:

1. Emu8086 to run the program.
2. Notepad to write the program.
3. 8086 microprocessor kit.

PROBLEM STATEMENT:

Solve the following series:-

(a) 5 + 6 + 8+...+ $(110)_{10}$ with help of jump instruction.
(b) 1 + 2 + 3+………………………………..+ $(101)_{10}$ with help of loop instruction.

Program1: Series 1

```
.MODEL SMALL
.CODE
MOV CX, 14
MOV AX, 5H
MOV BX, 6H
MOV SI, 1H
XXX: ADC AX, BX
INC SI
ADC BX, SI
DEC CX
JNZ XXX
END
```

Program2: Series 2

```
.MODEL SMALL
.CODE
MOV CX, 100
MOV AX, 1H
MOV BX, 2H
XXX: ADC AX, BX
INC BX
LOOP XXX
END
```

DATA TABLE:-

Registers	Value (After the last instruction)	Essential or not	Comments
AX			
BX			
CX			
DX			
BP			
SP			
SI			
DI			
CS			
DS			
ES			
SS			
IP			
F			

REPORT WRITING:-

1. Include experiment name, objectives, necessary software/ equipment and problem statement.
2. Write the program with brief explanation.
3. Insert the printed copy of “list file” of the program.
4. Find the result of all related registers from kit box for each line of the program.
5. Verify the series operation by theoretical calculation.
6. Finding the active flags from flag register and comparing them with the calculated values.
7. Write discussion including all the key points from the result of the program.
8. Write discussion if you have any error in the program or troubleshooting of the program.
9. Write the answer of the related questions given below.

RELATED QUESTIONS:-

Solve the following series:-

(a) 5 + 7 + 9+11+…... + $(115)_{10}$ with jump instruction.

(b) 4 + 6 + 10+16+… .... + $(114)_{10}$ with loop instruction.

List file

```
EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.exe_ -- emu8086 assembler version: 4.08

[ 10/18/2024  --  6:30:45 PM ]

==================================================================
==================================
[LINE]     LOC: MACHINE CODE                          SOURCE
==================================================================
==================================

[   1]        :                                 .MODEL SMALL
[   2]        :
[   3]        :                                 .CODE
[   4]    0000: B9 0E 00                            MOV CX, 14
[   5]    0003: B8 05 00                            MOV AX, 5H
[   6]    0006: BB 06 00                            MOV BX, 6H
[   7]    0009: BE 01 00                            MOV SI, 1H
[   8]    000C: 13 C3                             XXX: ADC AX, BX
[   9]    000E: 46                              INC SI
[  10]    000F: 13 DE                               ADC BX, SI
[  11]    0011: 49                              DEC CX
[  12]    0012: 75 F8                             JNZ XXX END
[  13]        :
[  14]        :
```

List file 2.

EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.exe_ -- emu8086 assembler version: 4.08

[10/18/2024 -- 6:34:37 PM]

```
===========================================================================================================
[LINE]     LOC: MACHINE CODE                          SOURCE
===========================================================================================================

[   1]        :                                   .MODEL SMALL
[   2]        :                                   .CODE
[   3]    0000: B9 64 00                                  MOV CX, 100
[   4]    0003: B8 01 00                                  MOV AX, 1H
[   5]    0006: BB 02 00                                  MOV BX, 2H
[   6]    0009: 13 C3                                   XXX: ADC AX, BX
[   7]    000B: 43                                    INC BX
[   8]    000C: E2 FB                                    LOOP XXX END
[   9]        :
[  10]        :

===========================================================================================================
```